

Chapter 4: Moving Charges and Magnetism

Formula Sheet + Tips & Tricks

Formula Sheet

Constants Used Throughout This Chapter

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1} \quad e = 1.6 \times 10^{-19} \text{ C}$$

$$m_e = 9.1 \times 10^{-31} \text{ kg} \quad m_p = 1.67 \times 10^{-27} \text{ kg}$$

$$m_\alpha = 6.7 \times 10^{-27} \text{ kg} \quad g = 9.8 \text{ m s}^{-2}$$

μ_0 = permeability of free space; e = elementary charge; m_e, m_p, m_α = rest masses of electron, proton, α -particle.

I. Magnetic Field due to Current (Biot–Savart & Ampere)

Field due to infinite straight wire

$$B = \frac{\mu_0 I}{2\pi r}$$

I = current, r = perpendicular distance from wire.

Field at centre of a circular coil

$$B = \frac{\mu_0 N I}{2r}$$

N = number of turns, r = radius of coil.

Field on the axis of a circular loop

$$B = \frac{\mu_0 I a^2}{2(a^2 + x^2)^{3/2}}$$

a = radius of loop, x = distance of point from centre along axis.

Field inside a long solenoid

$$B = \mu_0 n I$$

n = number of turns per unit length = N/L .

Field due to a moving point charge

$$B = \frac{\mu_0}{4\pi} \frac{qv \sin \theta}{r^2}$$

q = charge, v = speed, θ = angle between v and line joining charge to point.

II. Magnetic Force on a Moving Charge**Lorentz magnetic force**

$$F = qvB \sin \theta$$

q = charge, v = speed, B = field, θ = angle between $\vec{v}$ and $\vec{B}$.

III. Path of a Charged Particle in a Uniform Field**Radius of circular path**

$$r = \frac{mv}{qB}$$

m = mass, v = speed (perpendicular to B).

Time period & frequency of revolution

$$T = \frac{2\pi m}{qB}, \quad f = \frac{qB}{2\pi m}$$

Independent of the speed v of the particle.

Speed from accelerating potential

$$v = \sqrt{\frac{2qV}{m}}$$

V = accelerating potential difference.

Oblique entry (helical/spiral path)

$$r' = \frac{mv}{qB \sin \theta}$$

θ = angle between velocity and field; $\theta = 90^\circ$ gives circular path.

IV. Magnetic Force on a Current-Carrying Conductor**Force on a current-carrying conductor**

$$F = BIL \sin \theta$$

L = length of conductor, θ = angle between current direction and B .

Microscopic (drift velocity) form

$$I = nA v_d e$$

n = free electron density, A = area of cross-section, v_d = drift velocity.

V. Force between Parallel Straight Current-Carrying Conductors**Force per unit length between two wires**

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$$

d = separation between wires; like currents attract, unlike currents repel.

VI. Cyclotron**Cyclotron magnetic field**

$$B = \frac{2\pi m f}{q}$$

f = frequency of the applied oscillating voltage (= revolution frequency).

Maximum velocity & energy

$$v_{max} = 2\pi f R, \quad E_{max} = \frac{1}{2} m v_{max}^2$$

R = radius of the dees.

VII. Torque on a Current-Carrying Coil in a Magnetic Field**Torque on a current loop**

$$\tau = N B I A \sin \theta$$

A = area of coil, θ = angle between the normal to coil and $\vec{B}$.

Magnetic dipole moment

$$m = N I A$$

Torque may also be written $\tau = m B \sin \theta$; maximum torque $\tau_{max} = m B$ when $\theta = 90^\circ$.

VIII. Moving Coil Galvanometer: Sensitivities**Current sensitivity**

$$I_s = \frac{N B A}{k}$$

k = torsional constant of the suspension.

Voltage sensitivity

$$V_s = \frac{I_s}{R} = \frac{NBA}{kR}$$

R = resistance of the galvanometer coil.

IX. Conversion of Galvanometer into Ammeter/Voltmeter

Shunt resistance (galvanometer $\rightarrow$ ammeter)

$$S = \frac{I_g R_g}{I - I_g}$$

R_g = galvanometer resistance, I_g = full-scale deflection current, I = desired ammeter range.

Series resistance (galvanometer $\rightarrow$ voltmeter)

$$R = \frac{V}{I_g} - R_g$$

V = desired voltmeter range.

Net (effective) resistance

$$R_{ammeter} = \frac{R_g S}{R_g + S}, \quad R_{voltmeter} = R_g + R$$

Ammeter resistance is always very small; voltmeter resistance is always very large.

Tips, Tricks & Hacks

- Biot-Savart law gives $d\vec{B}$ due to a current element; always check the direction with the right-hand (screw) rule before plugging into the formula.
- For a straight wire, field $\propto \frac{1}{r}$ (not r^2) — don't confuse with the electric field/Coulomb pattern.
- For a circular loop, field at the centre is often asked to be compared with field on the axis ($B_{axis} = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$) — setting $x = 0$ must reduce to the centre formula; use this to check your derivation.
- Force on a moving charge: $\vec{F} = q\vec{v} \times \vec{B}$ — force is always $\perp$ to velocity, so it changes

direction, not speed $\Rightarrow$ path is a circle (or helix if there's a velocity component along $\vec{B}$).

- Radius of circular path $r = \frac{mv}{qB}$ and time period $T = \frac{2\pi m}{qB}$ — notice T is independent of speed, which is exactly why a cyclotron works at fixed frequency.
- Force between two parallel current-carrying wires: same direction currents attract, opposite directions repel — opposite to the “like charges repel” intuition from electrostatics, a classic trap.
- Moving coil galvanometer: current sensitivity $= \frac{NBA}{k}$, voltage sensitivity $= \frac{NBA}{kR}$ — increasing turns N boosts current sensitivity but also raises coil resistance, so voltage sensitivity may not increase proportionally.
- To convert galvanometer to ammeter, connect a small shunt in *parallel*; to convert to voltmeter, connect a large resistance in *series* — remembering “ammeter = parallel shunt, voltmeter = series resistor” avoids 90% of mistakes here.